

ECE 3210 hw05

1. Find the Fourier transform of

$$f(t) = 2 \operatorname{sinc}(\pi t) \cos(\pi t)$$

Solution:

$$\begin{aligned} f(t) &= 2 \operatorname{sinc}(\pi t) \cos(\pi t), \\ \operatorname{sinc}(\pi t) &= \frac{\pi}{\pi} \operatorname{sinc}(\pi t), \quad \operatorname{sinc}(\pi t) \longleftrightarrow \operatorname{rect}\left(\frac{\omega}{2\pi}\right). \end{aligned}$$

Modulating by $\cos(\pi t)$ shifts the spectrum by $\pm\pi$ and scales it by $1/2$:

$$\begin{aligned} F(\omega) &= 2 \cdot \frac{1}{2} \left[\operatorname{rect}\left(\frac{\omega - \pi}{2\pi}\right) + \operatorname{rect}\left(\frac{\omega + \pi}{2\pi}\right) \right] \\ &= \operatorname{rect}\left(\frac{\omega}{4\pi}\right). \end{aligned}$$

2. (a) Find the Fourier transform of

$$f(t) = \begin{cases} -t, & -2 \leq t \leq 0, \\ t, & 0 < t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Solution: Notice that

$$f(t) = \text{rect}\left(\frac{t}{4}\right) - 2\Delta\left(\frac{t}{4}\right).$$

Using the transform pairs for the rectangular and triangular pulses,

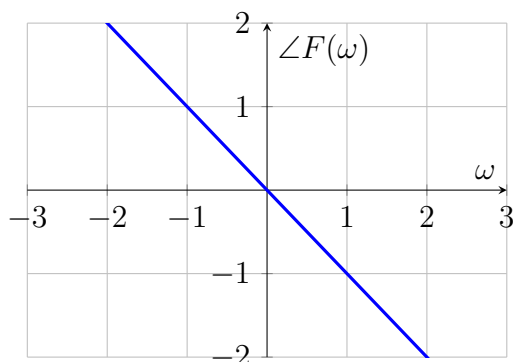
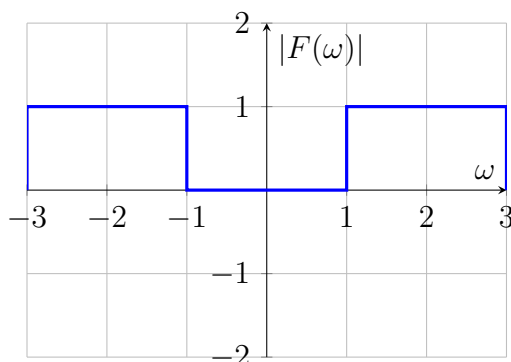
$$\text{rect}\left(\frac{t}{4}\right) \longleftrightarrow 4 \text{sinc}(2\omega),$$

$$\Delta\left(\frac{t}{4}\right) \longleftrightarrow 2 \text{sinc}^2(\omega),$$

so

$$F(\omega) = 8 \text{sinc}(2\omega) - 4 \text{sinc}^2(\omega).$$

- (b) Consider the magnitude and phase shown below. Find the inverse transform $f(t)$.



Solution: Set $\tilde{F}(\omega) = |F(\omega)|$ and apply the linear phase after finding $\tilde{f}(t)$:

$$\tilde{F}(\omega) = \text{rect}\left(\frac{\omega}{6}\right) - \text{rect}\left(\frac{\omega}{2}\right),$$

$$\text{rect}\left(\frac{\omega}{2}\right) \longleftrightarrow \frac{1}{\pi} \text{sinc}(t),$$

$$\text{rect}\left(\frac{\omega}{6}\right) \longleftrightarrow \frac{3}{\pi} \text{sinc}(3t).$$

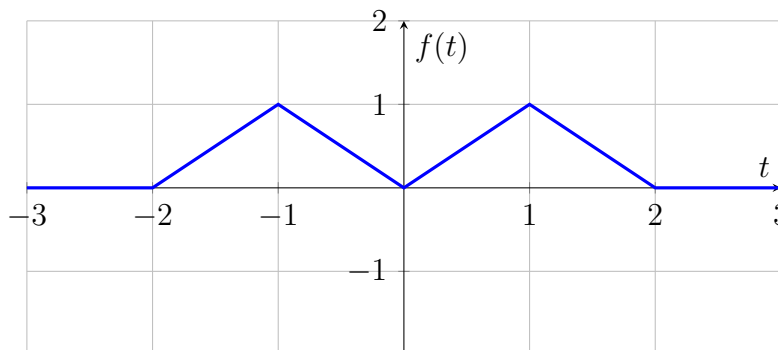
Thus

$$\tilde{f}(t) = \frac{1}{\pi} [3 \text{sinc}(3t) - \text{sinc}(t)].$$

Since $F(\omega) = \tilde{F}(\omega)e^{-j\omega}$,

$$f(t) = \tilde{f}(t-1) = \frac{1}{\pi} [3 \operatorname{sinc}(3(t-1)) - \operatorname{sinc}(t-1)] .$$

3. (a) Determine the Fourier transform of the waveform below.



Solution: Notice that

$$f(t) = 2\Delta\left(\frac{t}{4}\right) - 2\Delta\left(\frac{t}{2}\right).$$

The transform pairs give

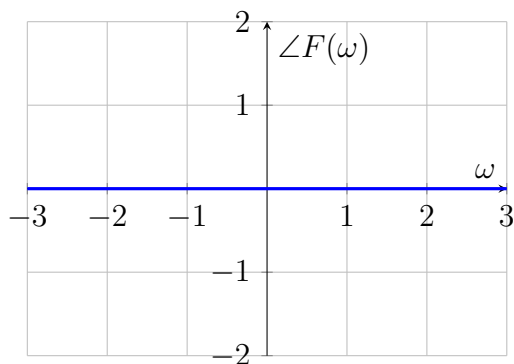
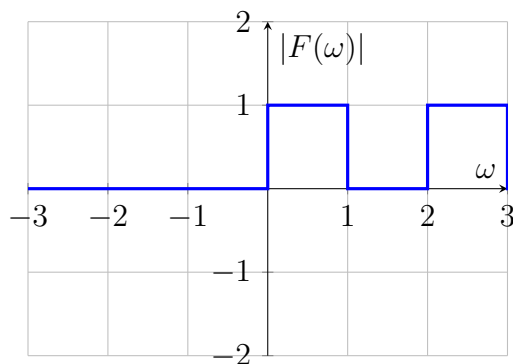
$$\Delta\left(\frac{t}{4}\right) \longleftrightarrow 2\text{sinc}^2(\omega),$$

$$\Delta\left(\frac{t}{2}\right) \longleftrightarrow \text{sinc}^2\left(\frac{\omega}{2}\right).$$

Therefore,

$$F(\omega) = 4\text{sinc}^2(\omega) - 2\text{sinc}^2\left(\frac{\omega}{2}\right).$$

- (b) The magnitude and phase of a spectrum are supplied below. Find $f(t)$.



Solution: Center the waveform by defining

$$\tilde{F}(\omega) = |F(\omega)| = \text{rect}\left(\frac{\omega}{3}\right) - \text{rect}(\omega).$$

Hence

$$\text{rect}\left(\frac{\omega}{3}\right) \longleftrightarrow \frac{3}{2\pi} \text{sinc}\left(\frac{3t}{2}\right),$$

$$\text{rect}(\omega) \longleftrightarrow \frac{1}{2\pi} \text{sinc}\left(\frac{t}{2}\right),$$

so

$$\tilde{f}(t) = \frac{1}{2\pi} \left[3 \text{sinc}\left(\frac{3t}{2}\right) - \text{sinc}\left(\frac{t}{2}\right) \right].$$

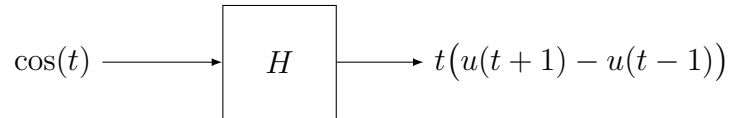
The phase gives $F(\omega) = \tilde{F}(\omega - 3/2)$, and therefore

$$f(t) = \tilde{f}(t)e^{j3t/2}$$

which is

$$f(t) = \frac{1}{2\pi} \left[3 \text{sinc}\left(\frac{3t}{2}\right) - \text{sinc}\left(\frac{t}{2}\right) \right] e^{j3t/2}.$$

4. (a) For the input/output relationship below, find $Y(\omega)$. Check the symmetry of your final answer.



Solution: Since

$$y(t) = t \operatorname{rect}\left(\frac{t}{2}\right),$$

differentiate the rectangular pulse:

$$y'(t) = \operatorname{rect}\left(\frac{t}{2}\right) - \delta(t+1) - \delta(t-1)$$

and take the Fourier transform

$$\begin{aligned} j\omega Y(\omega) &= 2 \operatorname{sinc}(\omega) - e^{j\omega} - e^{-j\omega} \\ &= 2 [\operatorname{sinc}(\omega) - \cos(\omega)]. \end{aligned}$$

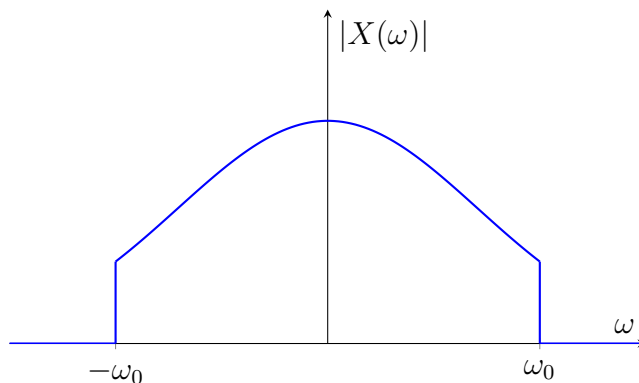
Thus

$$Y(\omega) = \frac{2}{j\omega} [\operatorname{sinc}(\omega) - \cos(\omega)].$$

- (b) Is the system in part (a) linear and time invariant?

Solution: The output creates new frequency information from the input, so the system is not LTI.

- (c) Let $x(t)$ be a signal whose spectrum is identically zero outside the range $-\omega_0 \leq \omega \leq \omega_0$. An example of such a spectrum is shown below. The actual values of $X(\omega)$ are not important for this problem, but the range of ω for which $X(\omega)$ is non-zero is.



For $y(t) = x(t) + x(t - 1)$, if $X(\omega) = 0$ outside $[-\omega_0, \omega_0]$, state the possible nonzero range of $Y(\omega)$.

Solution:

$$y(t) = x(t) + x(t - 1),$$

$$Y(\omega) = X(\omega) + X(\omega)e^{-j\omega}.$$

The multiplication by the phase factor does not change the support, so $-\omega_0 \leq \omega \leq \omega_0$.

(d) For $y(t) = x(t) \cos(\omega_0 t)$, if $X(\omega) = 0$ outside $[-\omega_0, \omega_0]$, state the possible nonzero range of $Y(\omega)$.

Solution:

$$y(t) = x(t) \cos(\omega_0 t),$$

$$Y(\omega) = \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)].$$

The two shifted copies occupy $-2\omega_0 \leq \omega \leq 2\omega_0$.

5. (a) Compute the integral

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1}{1 - j\omega} \right|^2 d\omega.$$

Solution: By Parseval, use $f(t) = e^t u(-t)$, whose transform is $F(\omega) = 1/(1 - j\omega)$:

$$\begin{aligned} E_Y &= \int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^0 e^{2t} dt \\ &= \frac{e^{2t}}{2} \Big|_{-\infty}^0 = \frac{1}{2}. \end{aligned}$$

- (b) Compute the integral

$$\int_{-\infty}^{\infty} \cos(2\pi t) \operatorname{sinc}^2(t) \operatorname{sinc}(2t) dt.$$

Solution: Let $f_1(t) = \operatorname{sinc}(2t) \cos(2\pi t)$. Then

$$\operatorname{sinc}(2t) \longleftrightarrow \frac{\pi}{2} \operatorname{rect}\left(\frac{\omega}{4}\right).$$

Modulation gives

$$F_1(\omega) = \frac{\pi}{4} \left[\operatorname{rect}\left(\frac{\omega - 2\pi}{4}\right) + \operatorname{rect}\left(\frac{\omega + 2\pi}{4}\right) \right].$$

Also $\operatorname{sinc}^2(t) \longleftrightarrow \pi \Delta(\omega/4)$, whose support is $[-2, 2]$. The supports of $F_1(\omega)$ and $\pi \Delta(\omega/4)$ do not overlap, so the generalized Parseval integral is zero:

$$\int_{-\infty}^{\infty} \cos(2\pi t) \operatorname{sinc}^2(t) \operatorname{sinc}(2t) dt = 0.$$

- (c) Evaluate the integral

$$I = \int_{-\infty}^{\infty} \frac{2\operatorname{sinc}(t)}{t^2 + 1} dt$$

Solution: Write

$$f_1(t) = \frac{2}{t^2 + 1}, \quad f_2(t) = \operatorname{sinc}(t).$$

By duality,

$$\frac{2}{t^2 + 1} \longleftrightarrow 2\pi e^{-|\omega|}, \quad \text{sinc}(t) \longleftrightarrow \pi \text{rect}\left(\frac{\omega}{2}\right).$$

Therefore Parseval gives

$$\begin{aligned} I &= \frac{1}{2\pi} \int_{-1}^1 (2\pi e^{-|\omega|})(\pi) d\omega \\ &= 2\pi \int_0^1 e^{-\omega} d\omega = 2\pi(1 - e^{-1}). \end{aligned}$$